



Neural network optimal control for discrete-time nonlinear systems with known internal dynamics

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Abstract

A neural network (NN) optimal control for discrete-time nonlinear dynamic systems with known internal dynamics is designed. The control is described by an algebraic equation with a variable structure. This algebraic equation is derived analytically. A functional block diagram of the controlled system is given and analyzed. Software and hardware implementation aspects of the controller are discussed. The controller does not need any training and has moderate complexity. The discrete-time state variable trajectories of the controlled system are shown to be globally asymptotically stable and convergent to unique steady states. It is proved that these trajectories converge to steady-state neighborhood in a finite number of steps. Sliding mode analysis of controller operation is fulfilled. A correctness of controller operation in the case of disturbances of its nonlinearities is analyzed. Using the controller for a special case of optimal tracking control is discussed. Results of presented computer simulations of optimal control of discrete-time two-dimensional and three-dimensional affine nonlinear systems and optimal tracking control of permanent-magnet motor of linear type applied for accurate positioning and nonlinear cooling continuous stirred tank reactor confirm theoretical statements of the paper and illustrate a performance of the controller.

Keywords Discrete time · Neural network · Nonlinear system · Optimal control

1 Introduction

Optimal control provides generating the most desirable output signal of a dynamic system in a certain time range such that the system performance is optimized. A solution of an optimal control problem can be found by a Pontryagin principle that gives a necessary condition of optimality or by obtaining a solution of the Hamilton–Jacobi–Bellman (HJB) equation that is a sufficient condition [1, 2]. An analytical solution of the HJB equation does not exist in the general multidimensional case. For this reason, the solution of the HJB equation is usually obtained by numerical methods. The solution of HJB equation can be obtained, e.g., using artificial neural networks (NNs) [3–5], reinforcement learning [6], approximate policy iteration [7], an expansion of Taylor series [8], splines [9] or Galerkin method [10].

A proportional integral derivative (PID) controller that uses an automatic control by iterative learning is described in [11]. The approach does not need a control loop for tuning, but it requires a signal of periodic reference. The controller iteratively adjusts the control signal using only the reference signal. A performance of the approach has been demonstrated based on control of affine nonlinear systems. The event-triggered decentralized control using adaptive critic designs is investigated in [12] to solve the HJB equations via local policy iteration algorithm. The single-critic networks are employed to achieve the approximation of optimal value functions. The Lyapunov stability proof ensures the asymptotic stability of the whole system and the uniform ultimate boundedness of the critic NNs. Finite-horizon approximate dynamic programming (ADP) algorithm is described leveraging an actor–critic framework in [13]. It is shown that the algorithm guarantees convergence to optimal solution. With the use of multilayer NNs in the actor–critic architecture, the algorithm is suitable to solve continuous-time finite-horizon optimal control problems for general nonlinear and

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complex systems. In [14], a control approach is described to solve optimal control problems of infinite horizon for continuous-time nonlinear systems that does not need a knowledge of the system mathematical model. Based on system inputs and outputs, a recurrent NN is used to identify the system dynamics. An ADP algorithm is designed to find the optimal control. The control error and disturbance of the system are used as the system control inputs. Single-layer NNs are applied as the networks of critic and actor for approximation of the performance and control, respectively. An approximate solution that is obtained in real-time using policy iteration for the problem of optimal control of infinite horizon for affine in the input nonlinear dynamic systems with known internal dynamics is described in [15]. The algorithm is designed in the form of actor/critic structure implementing simultaneous updating actor and critic NNs. The approach determines the optimal cost and control policy approximations in online mode and assures stability of the control operation. A NN for tracking control of continuous-time nonlinear dynamic systems is described in [16]. A first order differential equation with a variable structure and output equation are used to describe the network. A functional block diagram of the network is presented. The network is simple; it provides limited tracking error and convergence time to steady states and does not require any training. Nearly optimal control for the states of the quantum system is described in [17]. The quantum control optimization is fulfilled analytically within some protocol families that depend on a small set of parameters. Analytic solutions of optimal control problems with constraints are presented in [18]. The approach uses canonical duality theory to derive the analytic solution by transforming a constrained optimal control problem into a global optimization problem. An analytic optimal control approach for the virtually coupled train set for maintaining consistent speed and safe spacing between trains is described in [19]. A state-space model is firstly formulated to describe the virtually coupled train dynamics, based on which an optimal control is then designed using constraints of safe spacing, operation limits and train dynamic performance. To solve the constrained optimal control problem, an analytical algorithm is designed based on Pontryagin maximum principle.

Discrete-time systems are robust and provide a high accuracy of processing data and signals, especially in the presence of errors [20]. A multistep heuristic dynamic programming approach for solving the optimal control problem of nonlinear discrete-time systems is described in [21]. The approach speeds up value iteration and avoids the requirement of initial admissible control policy in policy iteration at the same time. The optimal control problem solution converges to the Bellman equation solution. For implementation purpose, the actor-critic NN structure is

developed. The critic NN is employed to estimate the value function and its NN weight vector is computed with a least-square scheme. The actor NN is used to estimate the control policy and a gradient descent method is applied for updating its NN weight vector. An optimal neural controller of inverse type for discrete-time nonlinear systems with uncertainties providing stabilization a required trajectory is presented in [22]. The controller uses the NN model that is trained in real time by a modified Kalman filter. A discrete-time Lyapunov function presented in quadratic form is minimized by the gradient algorithm to find the HJB equation solution. An approach based on policy iterative ADP is suggested to determine the optimal control of discrete-time nonlinear systems in [23]. It has been proved that the performance function converges to the optimal solution of the HJB equation. Stability of the nonlinear controlled systems is provided. NNs are employed for approximation of the performance function and to determine the optimal policy of the ADP approach. A NN structure to solve the optimal tracking problem of nonlinear discrete-time-varying system in online mode is presented in [24]. A multiple-model approach combined with adaptive self-organizing map NN is used to detect changes in the system dynamics. The number of sub-models is determined adaptively and grows once a mismatch between the stored sub-models and the new data is detected. Each sub-model contributes into the value function based on a responsibility signal obtained by the NN. Policy iteration and the value iteration algorithms are presented to find the optimal control for the partially unknown nonlinear discrete-time-varying systems in online mode.

The stability of nonlinear discrete-time systems controlled by optimal sequence of inputs minimizing infinite horizon discounted cost is presented in [25]. The stability is proved using the Lyapunov function. Uniform semi-global stability is ensured. The stability for discrete-time Takagi and Sugeno (T-S) fuzzy systems with perturbation disturbance and time delay is studied in [26]. Stochastic delay-dependent stability criteria are derived for stochastic T-S fuzzy systems with time-invariant and time-varying delays, respectively. The stability for the digital redesign of discrete-time switched systems using H_∞ linear matrix inequality is investigated in [27]. A discrete-time switching system with stable and unstable modes is proposed. H_∞ linear matrix inequality and Lyapunov functions are used to prove the stability.

Optimal control for discrete-time affine in the input nonlinear dynamic systems with the known internal dynamics is currently derived by numerical methods. This leads to high computational complexity of the solution that reduces its accuracy and increases time necessary to obtain the solution. Increasing accuracy and speed of the control

are important and actual problems in many applications, especially if it is required to fulfill a precise control in real time. The main contribution of this paper is that the NN optimal control is derived analytically and described by the algebraic equation with variable structure. Therefore, unlike other analogs it is not necessary to use numerical methods to obtain the optimal control. This allows to increase accuracy and reduce time needed to solve the optimal control problem. We believe that the analytical method described in this paper is used first time to find the control for affine in the input discrete-time nonlinear dynamic systems with the known internal dynamics. The controller has moderate complexity, and any tuning process is not required. A convergence analysis of the controlled system state variable trajectory to the steady states is fulfilled. The upper limit of the number of steps of the trajectory of the discrete-time state variable to the steady-state neighborhood is derived. Sliding mode analysis of the controller is fulfilled. Correctness of the controller operation in the case of disturbances of its nonlinearities is discussed. The computer simulations confirming theoretical statements and demonstrating performance of the controller are fulfilled.

The remainder of this paper is organized as follows: The problem formulation and its preliminaries are discussed in Sect. 2; In Sect. 3, the mathematical model and corresponding functional block diagram of the discrete-time NN optimal control are described. The existence and uniqueness of steady states, stability, and convergence analysis of trajectories of state variable to steady states, convergence quality of trajectories of state variable to steady states, sliding mode analysis, and controller nonlinearity disturbances are analyzed in Sect. 4. In Sect. 5, computer simulation results of the designed controller operation are presented. Concluding remarks are given in Sect. 6.

2 Problem formulation and preliminaries

Let us obtain a mathematical model of affine in the input, time-invariant discrete-time nonlinear dynamic system based on description of corresponding continuous-time system. For this purpose, we describe the discrete-time nonlinear dynamic system as follows [15]:

$$x(k+1) = x(k) + \tau(f(x(k)) + g(x(k))u(k)), x(1) = x_1, \quad (1)$$

where $x(k) \in \mathbb{R}^n$ is the discrete-time state variable, $f(x(k)) \in \mathbb{R}^n$ is the discrete-time nonlinear function of system internal dynamics, $g(x(k)) \in \mathbb{R}^{n \times m}$ is the discrete-time nonlinear function of system coupling dynamics,

$u(k) \in \mathbb{R}^m$ is the discrete-time control, $\tau = t(k+1) - t(k)$ is the discrete-time step, $k = 1, 2, \dots$

We assume that $f(0) = 0$ and the system can be stabilized on a set $u(k) \in \mathbb{R}^m$ that includes the origin, i.e., there exists such a discrete-time control function $u(k) \in U$ that the system is asymptotically stabilized on $\Omega \subseteq \mathbb{R}^n$. The functions of system internal dynamics $f(x(k))$ and $g(x(k))$ are supposedly known for any $k \in [1, k_{\max}]$, $k_{\max} < \infty$. We also suppose that the system is observable and controllable, i.e., $g(x(k)) \neq 0$. If $g(x(k)) = 0$, the system is uncontrolled and can be presented as follows:

$$x(k+1) = x(k) + \tau f(x(k)), x(1) = x_1. \quad (2)$$

The infinite horizon discrete-time value function associated with (1) is given by

$$V(x(k)) = \sum_k^{\infty} r(x(k), u(k)), \quad (3)$$

where $r(x(k), u(k)) = Q(x(k)) + u(k)^T R u(k)$, $Q(x(k))$ is positive definite function, i.e., $\forall x(k) \neq 0, Q(x(k)) > 0$ and $x(k) = 0 \Rightarrow Q(x(k)) = 0$, $R \in \mathbb{R}^{m \times m}$ is the symmetric positive definite matrix. The problem is to find the control providing minimization of the function (3) associated with (1).

The nonlinear Lyapunov equation of discrete-time is presented as follows:

$$r(x(k), u(x(k))) + V_x^T(f(x(k)) + g(x(k))u(x(k))) = 0, \quad (4)$$

where $V_x = \Delta V / \Delta x$ is the gradient of value function (3), as the value function is not dependent explicitly on discrete-time step k . A Hamiltonian of the problem is given by

$$H(x(k), u(k), V_x) = r(x(k), u(k)) + V_x^T(f(x(k)) + g(x(k))u(k)). \quad (5)$$

Supposing existence and uniqueness of the cost function (3) minimum, the optimal control function can be presented as follows:

$$u(x(k)) = -\frac{1}{2}R^{-1}g^T(x(k))\bar{V}_x, \quad (6)$$

where $\bar{V}_x$ is the gradient of the optimal value function.

If we replace $u(x(k))$ in (4) with (6), then the HJB equation in terms of $\bar{V}_x$ can be given by

$$Q(x(k)) + \bar{V}_x^T f(x(k)) - \frac{1}{4}\bar{V}_x^T g(x(k))R^{-1}g^T(x(k))\bar{V}_x = 0, \quad (7)$$

where $\bar{V}(x(k))$ is the optimal value function that is presented as follows:

$$\bar{V}(x(k)) = \min_{u(k)} \sum_k^{\infty} r(x(k), u(k)), \quad (8)$$

$\bar{V}(1) = 0$. To determine the solution of the optimal control problem, the algebraic Eq. (7) should be solved relatively to $\bar{V}_x$ and the obtained solution substituted in (6). However, solving the nonlinear Eq. (7) in a multidimensional case is very hard or even impossible [15]. For this reason, approximate solutions of the optimal control problem are obtained using numerical methods. For instance, to obtain the solution of Eq. (7) one can approximate the value function by multilayer NN based on the theorem of Stone–Weierstrass [28]. However, an approximation error in this case approaches zero only if a quantity of the neurons and layers of the network is increased to infinity. This can lead to too high time and space complexity of the network.

3 The neural network optimal control of discrete-time

The continuous-time optimal control for affine in the input scalar nonlinear systems with the known internal dynamics derived analytically has been presented in [29]. Let us design the discrete-time NN optimal control for affine in the input nonlinear systems with the known internal dynamics. For this purpose, we use the affine in the input 1-D nonlinear dynamic system that is described by (1) with $m = n = 1$, i.e., $\bar{V}_x \ x(k) \in \mathbb{R}$, $f(x(k)) \in \mathbb{R}$, $g(x(k)) \in \mathbb{R}$, $u(k) \in \mathbb{R}$. The optimal control function in this case can be given by

$$u(x(k)) = -\frac{1}{2}R^{-1}g(x(k))\bar{V}_x. \quad (9)$$

The corresponding HJB equation can be presented as follows:

$$Q(x(k)) + \bar{V}(x(k)) - \frac{1}{4}(g(k))^2R^{-1}V_x^2 = 0. \quad (10)$$

Let us obtain optimal control by finding the solution $\bar{V}_x$ of the Eq. (10) and substituting this solution in (9). As one can see, (10) is a quadratic algebraic equation relatively to $\bar{V}_x$. Two possible solutions of this equation can be found in analytic way by the quadratic formula. Using such solutions and taking into account a convex form of the function (3) with its minimum at $x(k) = 0$, and that $f(1) = 0$ and $x(k) = 0 \Rightarrow Q(x(k)) = 0$, the gradient of optimal value function can be given by

$$\bar{V}_x = \frac{\Delta \bar{V}}{\Delta x(k)} = \frac{2}{R^{-1}(g(x(k)))^2}(f(x) + h(x(k))S(x(k))), \quad (11)$$

where

$$S(x(k)) = \begin{cases} 1, & \text{if } x(k) \geq 0; \\ -1, & \text{if } x(k) < 0 \end{cases} \quad (12)$$

is the scaled and shifted down threshold (Heaviside) activation function [29, 30] of discrete-time and

$$h(x(k)) = \sqrt{(f(x(k)))^2 + R^{-1}Q(x(k))(g(x(k)))^2}. \quad (13)$$

Replacing $\bar{V}_x$ in (9) with (11) leads to obtaining the control function

$$u(x(k)) = \frac{1}{g(x(k))}(f(x(k)) + h(x(k))S(x(k))). \quad (14)$$

Having replaced $u(x(k))$ in 1-D case of (1) with (14), we obtain the following difference equation that describes the controlled nonlinear system:

$$x(k+1) = x(k) - \tau h(x(k))S(x(k)). \quad (15)$$

For multidimensional system, the set of corresponding difference equations is given by

$$x_l(k+1) = x_l(k) - \tau_l h_l(x_l(k))S(x_l(k)), \quad l = 1, \dots, n, \quad (16)$$

$$\text{where } x_l(k) \in \mathbb{R}, \quad \tau_l \in \mathbb{R}, \quad h_l(x_l(k)) = \sqrt{(f_l(x_l(k)))^2 + R_l^{-1}Q_l(x_l(k))(g_l(x_l(k)))^2},$$

$$S(x_l(k)) = \begin{cases} 1, & \text{if } x_l(k) \geq 0; \\ -1, & \text{if } x_l(k) < 0, \end{cases} \quad f_l(x_l(k)) \in \mathbb{R}, R_l \in \mathbb{R}, Q_l(x_l(k)) \in \mathbb{R}, g_l(x_l(k)) \in \mathbb{R}$$

. The difference Eqs. (16) describe n -D controlled nonlinear system of discrete-time. The gradient of optimal value function (11) in this case is presented as follows:

$$\bar{V}_{x_l} = \frac{\Delta \bar{V}}{\Delta x_l(k)} = \frac{2}{R_l^{-1}(g_l(x_l(k)))^2}(f_l(x_l(k)) + h_l(x_l(k))S_l(x_l(k))), \quad l = 1, \dots, n. \quad (17)$$

The control function (14) is given by

$$u_l(x_l(k)) = \frac{1}{g_l(x_l(k))}(f_l(x_l(k)) + h_l(x_l(k))S_l(x_l(k))), \quad l = 1, \dots, n. \quad (18)$$

A functional block diagram of controlled nonlinear dynamic system described by the difference Eq. (15) is presented in Fig. 1. In particular, the controller consists of discrete-time blocks of summation $\sum$, multiplication $\cdot$, division $/$, square root extraction $\sqrt{\cdot}$, activation function $S(x(k))$, function $Q(x(k))$, and source of constant signals R^{-1} . As one can see in Fig. 1, the controller has two adders, three multipliers, one divider, one nonlinear function block (or no nonlinear function block if $Q(x(k)) = \text{const}$), one square root block, and one source of

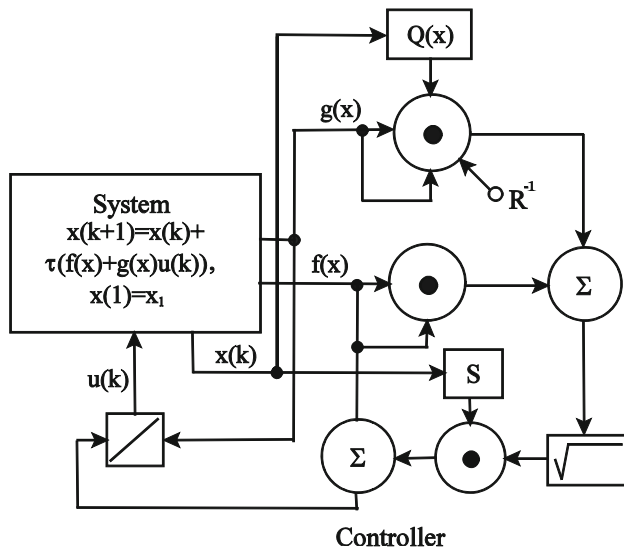


Fig. 1 The functional block diagram of the controlled nonlinear dynamic system of discrete-time described by the difference Eq. (15)

constant signals (or two sources of constant signals if $Q(x(k)) = \text{const}$).

Note that the controller is capable of operating correctly with any finite initial value of state variable $-\infty < x(1) < \infty$. For this reason, it does not require a repetitive resetting if parameters of the system are changed. An extra control circuit and spending additional time for such resetting are not necessary as well.

In the multidimensional case of discrete-time nonlinear systems described by the difference Eqs. (16), the functional block diagram shown in Fig. 1 presents l th, $l = 1, 2, \dots, n$ controlled discrete-time nonlinear dynamic system.

The controller shown in Fig. 1 is suitable for implementation in software. The software implementation can be chosen if simple, flexible, and accurate realization for different applications is required. In this case, e.g., Matlab, C++, Python or Compute Unified Device Architecture can be used [31]. However, in this case operation speed of the controller can be insufficient for real-time applications. Moreover, its miniaturization level or energy efficiency can be not high enough for practical applications. Therefore, the controller can be implemented in a digital hardware, if high operation speed, integration density or energy efficient realization are required [32–34].

The controller presented in Fig. 1 is simpler than other analogs since it does not have critic or actor parts [15]. The controller does not require solving optimization problem to determine the optimal control since such control is derived analytically. This reduces time and space complexity of the optimal control problem solution.

4 Analysis of the controlled nonlinear dynamic system

4.1 Existence and uniqueness of steady states

Let us analyze the existence and uniqueness of steady-state solutions of the difference Eq. (15). For this purpose, formulate the following theorem.

Theorem 1 *The steady-state solutions of difference Eq. (15) exist, and they are unique for any discrete-time step $0 < \tau < \infty$.*

Proof The equilibrium equation related to the difference Eq. (15) is given by

$$(\bar{x}(k+1) - \bar{x}(k))/\tau = h(\bar{x}(k))S(\bar{x}(k)) = 0. \quad (19)$$

According to (13), $h(\bar{x}(k)) = 0$ if $f(0) = 0$ and $Q(\bar{x}(k)) = 0$ since we supposed that $R > 0$ and $g(\bar{x}(k)) \neq 0$. We also assumed that $f(0) = 0$ and $\bar{x}(k) = 0 \Rightarrow Q(\bar{x}(k)) = 0$. Therefore, $h(\bar{x}(k)) = 0$ only if $f(0) = 0$ and $Q(\bar{x}(k)) = 0$ that leads to satisfying equilibrium Eq. (19) for any discrete-time step $0 < \tau < \infty$. This means that the steady-state solution of the difference Eq. (15) exists; it is unique and equal to zero for arbitrary $0 < \tau < \infty$.

4.2 Stability and convergence analysis of state variable trajectories to steady states

Since the difference Eq. (15) describes a closed-loop discrete-time nonlinear dynamic system, it is required that this system should be stable. This means that the discrete-time state variable trajectory $x(k)$ should start from the initial state $x(1)$ and after convergence reach steady-state value $x(k) = \bar{x}(k) = 0$. To guarantee the stability of difference Eq. (15) solutions it is necessary to fulfill a convergence analysis of such solutions to steady states. Let us use for this purpose the second method of Lyapunov [35]. Derive sufficient conditions of global asymptotic stability and convergence of Eq. (15) solutions to steady states. The availability of such conditions states the following lemma.

Lemma 1 *For arbitrary initial value $-\infty < x(1) < \infty$, discrete-time step $0 < \tau < 2|x(k)|/h(x(k))$, $Q(x(k)) \geq 0$, $R > 0$, and any finite $Q(x(k)) \geq 0$, the solution trajectories of (15) are globally asymptotically Lyapunov stable and globally converge to steady states.*

Proof By shift of the origin to the equilibrium point $\bar{x}(k)$ the difference Eq. (15) can be presented in the equivalent Persidskii type form [35] that is the same as (15) since $\bar{x}(k) = 0$ and $f(\bar{x}(k)) = 0$, $Q(\bar{x}(k)) = 0$. Therefore, it follows from (13) that $h(\bar{x}(k)) = 0$ leading to $\bar{x}(k+1) = 0$ according to (15). A candidate of the Lyapunov function associated with the system (15) can be given by

$$W(k) = |x(k)|. \quad (20)$$

As one can see, (20) is the convex bounded from below discrete-time function. Moreover, $W(k) > 0$ if $x(k) \neq 0$ and $W(k) = 0$ if $x(k) = 0$. Consider the following difference:

$$\Delta W(k) = W(k+1) - W(k) = |x(k+1)| - |x(k)|. \quad (21)$$

Using the difference Eq. (15), difference (21) and the second method of Lyapunov one can write the following inequality:

$$\Delta W(k) = |x(k) - \tau h(x(k))S(x(k))| - |x(k)| \leq 0. \quad (22)$$

The inequality (22) can be given by

$$-|x(k)| \leq x(k) - \tau h(x(k))S(x(k)) \leq |x(k)|. \quad (23)$$

Consider the case $x(k) > 0$. In this case, using (12) one can present (23) as follows:

$$-x(k) < x(k) - \tau h(x(k)) < x(k). \quad (24)$$

Subtracting from (24) $x(k)$ yields

$$-2x(k) < -\tau h(x(k)) < 0. \quad (25)$$

Dividing (25) by $-h(x(k))$ gives $0 < \tau < 2x(k)/h(x(k))$.

Consider the second case $x(k) < 0$. In this case, using (12) inequalities (23) can be given by

$$x(k) < x(k) - \tau h(x(k)) < -x(k). \quad (26)$$

Subtracting from (26) $x(k)$ gives

$$0 < \tau h(x(k)) < -2x(k). \quad (27)$$

Dividing (27) by $h(x(k))$ one obtains $0 < \tau < -2x(k)/h(x(k))$.

Consider the third case $x(k) = 0$. In this case, using (22) one obtains $-\infty < \tau < \infty$.

Combining the three results obtained above yields the bounds $0 < \tau < 2|x(k)|/h(x(k))$. This means that $\Delta W(k) \leq 0$ for any $0 < \tau < 2|x(k)|/h(x(k))$. Therefore, it follows that $W(k)$ is decreasing positive discrete-time function for any $0 < \tau < 2|x(k)|/h(x(k))$. For this reason, if inequalities $0 < \tau < 2|x(k)|/h(x(k))$ are met, then $W(k) > 0 \forall x(k) \neq 0$ and $W(k) \equiv 0$ if $x(k) = 0$. Note that $W(k) \equiv 0$ implies that $x(k) \equiv 0$. In other words, if in some change range of k , the equality $x(k) \equiv 0$ meets, then $W(k) \equiv 0$ in such range. This implies that if $0 < \tau < 2|x(k)|/h(x(k))$, then (21) is a negative discrete-time function of $x(k)$. The function $W(k)$ is radially unbounded (i.e., $W(k) \rightarrow \infty$ as $|x(k)| \rightarrow +\infty$), bounded from below, it is convex, and its minimal value is equal to zero. For this reason, according to the stability theory of Lyapunov this function and the difference $\Delta x(k)$ converge to zero values. This guarantees that $x(k) = \bar{x}(k) = 0$ is globally asymptotically stable that means availability of globally attractive origin and globally asymptotically stable system state dynamics. At the

equilibrium point $x(k) = \bar{x}(k) = 0$, the function (20) accepts its smallest value equal to zero $\bar{W}(k) = 0$.

Thus, for bounded from below Lyapunov function (20), i.e., for $W(k) \geq 0$ with $W(k) = 0$ for $x(k) = 0$, if the solutions of difference Eq. (15) are changed, satisfying bounds $0 < \tau < 2|x(k)|/h(x(k))$ is a sufficient condition of global asymptotic Lyapunov stability and convergence for the trajectory of $x(k)$ to the steady-state value equal to zero.

An example of globally asymptotically stable and globally asymptotically convergent to zero state trajectory of the controlled discrete-time nonlinear dynamic system described by the difference Eq. (15) and corresponding control function (14) is shown in Fig. 2.

4.3 Convergence quality of state variable trajectories to steady states

Let us define a certain set

$$\Phi := \{\bar{x}(k) : W(\bar{x}(k)) = 0\}, \quad k = 1, \dots, m. \quad (28)$$

The difference Eq. (15) has a constant and equal to zero right-hand side for each Φ . A steady-state solution of this equation is such scalar $x(m) \in \mathbb{R}$ that $x(m) \in \Phi$. Let us analyze a quality of convergence to the steady states of solution trajectories of Eq. (15). For this purpose, we state and prove the following lemma.

Lemma 2 *The solution trajectories of the difference Eq. (15) converge, from any initial state $-\infty < x(1) < \infty$, to an arbitrary small neighborhood $\delta > 0$ of Φ (28) in a*

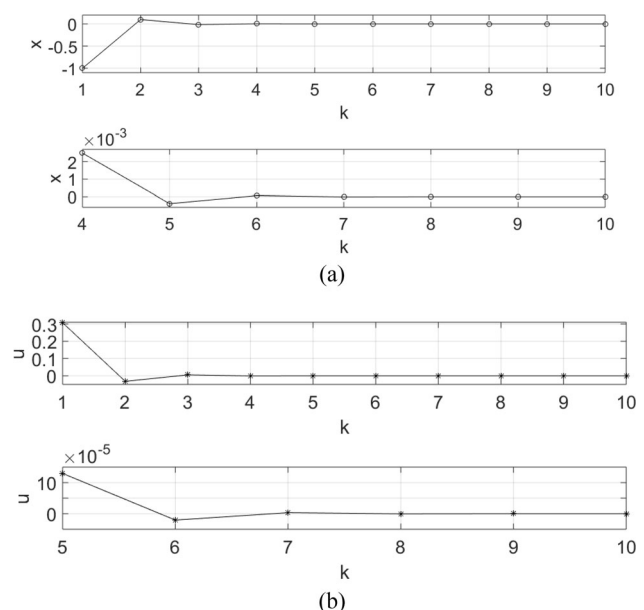


Fig. 2 **a** An example of state trajectory of the controlled discrete-time nonlinear dynamic system described by the difference Eq. (15); **b** an example of control function (14) trajectory

finite number m of steps k and remain in this neighborhood thereafter if $0 < \tau < 2|x(k)|/h(x(k))$.

Proof Consider the difference Eq. (15), difference (21) of (20) and the set (28). The aim is to show that there exists such a scalar $\delta > 0$ that for each $|x(k)| \geq \delta$ the difference $\Delta W(x(k)) \leq -\sigma$, $\sigma > 0$ and that Φ is an invariant set. This means that the state variable $x(k)$ trajectories converge to the steady-state set Φ neighborhood $\delta > 0$ in a limited number of steps and that the trajectories remain in δ implying that Φ is an invariant set.

Using Eq. (15), the difference (21) can be given by

$$\Delta W(x(k)) = |x(k) - \tau h(x(k))S(x(k))| - |x(k)|. \quad (29)$$

Taking into account (12), (29) can be presented as follows:

$$\begin{aligned} \Delta W(x(k)) &= |x(k) - \tau h(x(k))S(x(k))| - |x(k)| \\ &= |x(k) - \tau h(x(k))| - |x(k)| = -\tau h(x(k)). \end{aligned} \quad (30)$$

Based on (30), one obtains for each $|x(k)| \geq \delta$ that

$$\Delta W(x(k)) \leq -\tau h(\delta) = -\sigma. \quad (31)$$

To determine the limit for the number of steps required for reaching the neighborhood δ , the following bound can be used:

$$W(x(m)) \leq W(x(1)) - \sigma(m-1). \quad (32)$$

Using (32), the number of steps for $x(k)$ needed to reach the neighborhood δ is given by

$$m \leq (W(x(1)) - W(x(m)))/\sigma + 1 = (x(1) - \delta)/\sigma + 1. \quad (33)$$

Since $x(1)$ is limited, m is also limited for any $\delta > 0$ and $\sigma > 0$. Hence, $W(x(k))$ approaches the neighborhood σ in a limited number of steps. This means that the trajectories of solutions of Eq. (15) approach the neighborhood δ in a finite number of steps.

Taking into account (31) and (33), a time step to reach the neighborhood δ in m iterations can be determined as follows:

$$\tau \geq (x(1) - \delta)/((m-1)h(\delta)). \quad (34)$$

Let us show that the solution trajectories of Eq. (15) stay in the steady state Φ neighborhood $\delta > 0$ meaning that Φ is an invariable set. If $x(k) \in \Phi$, then $W(k) = |x(k)| \leq \delta$ and $\Delta W(x(k)) \leq 0$ by (22). If for some $x = K$, the variable $x(K)$ comes out of Φ , then $\Delta W(x(k)) < 0$ and $W(K) > 0$ that contradicts the fact that W is a non-increasing function along the trajectories of Eq. (15) solutions. For this reason, we can claim that the trajectories of difference Eq. (15) solutions approach the neighborhood $\delta > 0$ of Φ and remain in it thereafter.

4.4 Sliding mode analysis

The difference Eq. (15) has variable structure. A sign of finite difference $\Delta x(k) = x(k+1) - x(k)$ of (15) can be changed at the set Φ since its right-hand side has the discrete-time threshold activation function. Solutions of (15) are bounded by the set Φ . If the inequalities $0 < \tau < \infty$ hold and sliding mode conditions are met, a sliding motion can occur. This means that the discrete-time state variable of the controlled system can fulfill sliding motion along the set Φ . Let us suppose that the sliding mode equation is $x(k) = 0$. A sliding motion of Eq. (15) solutions at hyperplane $x(k)$ can occur if the following inequalities are satisfied [36]:

$$\Delta x(k) > 0, \Delta x(k) < 0. \quad (35)$$

$x(k) \rightarrow -0 \quad x(k) \rightarrow +0$

The inequalities (35) mean that if the discrete-time state variable $x(k)$ accepts a small negative value, its gradient $\Delta x(k)$ should be positive. This will force $x(k)$ to increase its value. On the other hand, if the discrete-time state variable $x(k)$ accepts a small positive value, its gradient $\Delta x(k)$ should be negative. For this reason, a value of $x(k)$ will be decreased.

Let us firstly analyze the controlled nonlinear system (15) in the case if $x(k) \rightarrow -0$. In this case, the sliding mode conditions (35) are satisfied since according to (12) and (15) $\Delta x(k) > 0$, if $x(k) < 0$. In another case, if $x(k) \rightarrow +0$ the conditions (35) also hold since in accordance with (12) and (15) $\Delta x(k) < 0$, if $x(k) > 0$. Thus, the sliding motion in the solutions of the controlled system described by (15) can occur since the conditions (35) of existence of sliding motion are satisfied.

Note that the limit $\Delta x(k) = 0$ means a possibility of the degenerative trajectory of Eq. (15) solutions. In this a case, the solutions of (15) are defined on the discontinuity surface Φ . If solutions of (15) have deviations, the degenerative movement can be converted to a sliding motion.

In real applications, the speed of the switching operation is limited. If the controlled system described by the difference Eq. (15) is implemented in real software or hardware, the errors of computations such as summation, multiplication, integration, and time delays can cause switching of Eq. (15) solutions instead of sliding along the set Φ . For this reason, the solutions of Eq. (15) can have limited frequency and amplitude oscillations [36]. An example of sliding motion in the form of decreasing oscillations in the solutions of the controlled discrete-time nonlinear dynamic system described by the difference Eq. (15) is shown in Fig. 2.

Notice that in real-world applications, the activation function (12) can be replaced with dead zone hard-limiting activation function given by

$$S(x(k)) = \begin{cases} 1, & \text{if } x(k) > \delta; \\ |1, 1|, & \text{if } -\delta \leq x(k) \leq \delta; \\ -1, & \text{if } x(k) < -\delta, \end{cases} \quad (36)$$

where $\delta > 0$ [20].

4.5 Disturbances of controller nonlinearities

The difference Eq. (15) has nonlinear functions (12) and (13). Such nonlinearities can be computed with inaccuracies which cause arising disturbances around their correct values. This in turn can generate deviations of Eq. (15) solutions. For this reason, conditions of robustness to deviations of the Eq. (15) solutions should be derived.

Let us suppose that (12) and (13) are not accurate functions but they are changed in a certain range. Determine the largest possible disturbances of such nonlinearities for which the functions (12) and (13) still operate correctly. For this purpose, the difference Eq. (15) with the disturbed nonlinearities can be presented as follows:

$$\tilde{x}(k+1) = \tilde{x}(k) - \tau \tilde{h}(\tilde{x}(k)) \tilde{S}(\tilde{x}(k)), \quad (37)$$

where

$$\tilde{S}(\tilde{x}(k)) = \begin{cases} 1 + \lambda, & \text{if } \tilde{x}(k) \geq 0; \\ -1 + \lambda, & \text{if } \tilde{x}(k) < 0, \end{cases} \quad (38)$$

is the disturbed threshold activation function (12),

$$\tilde{h}(\tilde{x}(k)) = \sqrt{(f(\tilde{x}(k)))^2 + R^{-1}Q(\tilde{x}(k))(g(\tilde{x}(k)))^2} + \mu \quad (39)$$

is the disturbed nonlinear function (13), $\tilde{x}(k)$ is the deviated state variable $x(k)$, λ and μ are the random noises with normal distribution, zero mean and standard deviations $\rho \geq 0$ and $\gamma \geq 0$ correspondingly.

Let us find limitations for the allowed disturbances of the activation function (12) for which it still operated properly. In order to fulfill correct rounding, largest changes of (12) have to be bounded. Considering (38) for a worst-case scenario, one can write the limit from above $\rho < 1$ that leads to the following bounds:

$$0 \leq \rho < 1. \quad (40)$$

Thus, for a correct computing (12), its disturbances should be limited by inequalities (40).

Let us determine bounds for the allowed disturbance of nonlinear function (13) for which this function is still computed with sufficient accuracy. To guarantee a positive value of the function $\tilde{h}(\tilde{x})$, its largest disturbances should be limited. For (39) and a worst-case scenario, one can write the inequalities given by

$$0 \leq \gamma < \sqrt{(f(\tilde{x}))^2 + R^{-1}Q(\tilde{x})(x(\tilde{x}))^2}. \quad (41)$$

This means that to guarantee a positive value of $\tilde{h}(\tilde{x})$ the disturbances of (39) should be constrained by inequalities (41).

Thus, it can be concluded that to obtain accurate solutions of Eq. (15), the disturbances of activation function (12) should satisfy the inequalities (40) and the disturbances of nonlinear function (13) should meet the bounds (41).

Note that the optimal control approach presented above for one-dimensional nonlinear dynamic system described by the difference Eq. (15) can be also applied for the special case of optimal tracking control, where $|r(k+1) - r(k)| < |y(k+1) - y(k)|$, $y(k)$ is the augmented discrete-time state variable, $r(k)$ is the discrete-time reference, $y(k) - r(k) = e(k)$ is the tracking error. In such case, the discrete-time state variable $x(k)$ of difference Eq. (15) can be replaced with $e(k)$, i.e., $x(k) = e(k)$ and the discrete-time state variable $x(k+1)$ of difference Eq. (15) can be replaced with $e(k+1)$, i.e., $x(k+1) = e(k+1)$.

Notice that the results presented above are also valid for n scalar controlled time invariant, affine in the input discrete-time nonlinear dynamic systems described by the set of difference Eqs. (16). In particular, the optimal control approach outlined above can be used in special case of optimal tracking control of nonlinear dynamic systems described by the set of difference Eqs. (16), where $|r_l(k+1) - r_l(k)| < |y_l(k+1) - y_l(k)|$, $l = 1, \dots, n$, $y_l(k)$ is the l th augmented discrete-time state variable, $r_l(k)$ is the l th discrete-time reference, $y_l(k) - r_l(k) = e_l(k)$ is the l th tracking error. In this case, the l th discrete-time state variable $x_l(k)$ can be replaced with $e_l(k)$, i.e., $x_l(k) = e_l(k)$ and the l th discrete-time state variable $x_l(k+1)$ can be replaced with $e_l(k+1)$, i.e., $x_l(k+1) = e_l(k+1)$.

5 Computer simulations of controller operation

Let us demonstrate an operation of the discrete-time affine in the input controlled nonlinear dynamic systems with the known internal dynamics described by the set of difference Eqs. (16). For this purpose, consider the following computer simulation examples.

Example 1 We use the input data described below: $n = 3$, $f_1(x_1(k)) = x_1^3(k)$, $f_2(x_2(k)) = -x_2^3(k)$, $f_3(x_3(k)) = -x_3^2(k)$, $f_4(x_4(k)) = -x_4^2(k)$, $f_5(x_5(k)) = x_5^2(k)$, $g_1(x_1(k)) = 1.5 + \cos(x_1(k))$, $g_2(x_2(k)) = 0.5 - \sin(x_2(k))$, $g_3(x_3(k)) = 2.5 - \cos(x_3(k))$, $g_4(x_4(k)) = 3.2 + \sin(x_4(k))$, $g_5(x_5(k)) = 4.7 + \cos(x_5(k))$, $Q_1(x_1(k)) = 1.7x_1^3(k)$, $Q_2(x_2(k)) = 1.4x_2^3(k)$, $Q_3(x_3(k)) = 3.1x_3^3(k)$, $Q_4(x_4(k)) = 2.5x_4^2(k)$, $Q_5(x_5(k)) = 4.3x_5^3(k)$, $R_1 = 2$, $R_2 = 4$, $R_3 = 6$, $R_4 = 1$,

$R_5 = 3, \tau = 0.04$ and initial conditions $x_1(1) = 2, x_2(1) = -4, x_3(1) = 8, x_4(1) = -3, x_5(1) = 5$. To simulate the controller operation in the presence of disturbances of its nonlinearities, we add the uniformly distributed in the open interval $(-0.01, 0.01)$ random noise to the right-hand sides of (16).

The trajectories of states $x_1(k), x_2(k), x_3(k), x_4(k)$ and $x_5(k)$, the dynamics of value function gradients $\Delta V/\Delta x_1, \Delta V/\Delta x_2, \Delta V/\Delta x_3, \Delta V/\Delta x_4$ and $\Delta V/\Delta x_5$, and the dynamics of control functions $u_1(k), u_2(k), u_3(k), u_4(k)$ and $u_5(k)$ are shown in Fig. 3. According to this figure, the trajectories of $x_1(k), x_2(k), x_3(k), x_4(k)$ and $x_5(k)$, gradients $\Delta V/\Delta x_1, \Delta V/\Delta x_2, \Delta V/\Delta x_3, \Delta V/\Delta x_4$ and $\Delta V/\Delta x_5$, and control functions $u_1(k), u_2(k), u_3(k), u_4(k)$ and $u_5(k)$ have globally asymptotically stable trajectories which are convergent to zero as needed.

Example 2 Compare the controller operation presented above with that based on methods of neural dynamic programming that are used to determine solution of the HJB equation in online mode to fulfill optimal control of general affine discrete-time nonlinear systems described in [37]. The techniques require approximation of actor's and critic's NNs. For this purpose, we employ affine in the input nonlinear dynamic system presented in [37] that is described by (1), where $m = n = 2, g(x(k)) = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$,

$f(x(k)) = \begin{bmatrix} x_1(k)x_2(k) - x_1(k) \\ x_1^2(k) + 1.8x_2(k) \end{bmatrix}$, $x_1(1) = x_2(1) = 0.5$. We set for HJB equation $Q(x) = x^T(k)x(k)$ and $R_1 = R_2 = 1$. In this case, we describe the state variable $x_1(k)$ by the first difference equation of (1) $x_1(k+1) = \tau x_1(k)x_2(k)$, where the discrete-time step is set to be $\tau = 1$. We consider this difference equation as one-dimensional with only one state variable $x_1(k)$ since $x_2(k)$ is determined from the second difference equation of (1) in each discrete-time point k . Note that since $g_1(x(k)) = 0$, the first difference equation

of (1) is a partial case of the difference Eq. (2). The second difference equation of (1) is also considered as one-dimensional with only one state variable $x_2(k)$ since $x_1(k)$ is determined from the first difference equation of (1) in each discrete-time point k . Therefore, the state variable $x_2(k)$ is described by the second difference equation of set (16), i.e., $x_2(k+1)$.

$= x_2(k) - \tau h_2(x_2(k))S(x_2(k))$, where $h_2(x_2(k)) = \sqrt{(f_2(x_2(k)))^2 + R^{-1}Q(x_2(k))(g_2(x_2(k))))^2}$, $S(x_2(k)) = \begin{cases} 1, & \text{if } x_2(k) \geq 0; \\ -1, & \text{if } x_2(k) < 0, \end{cases}$ $f_2(x_2(k)) = x_1^2(k) + 1.8x_2(k)$, $Q(x_2(k)) = x_1^2(k) + x_2^2(k)$, $g_2(x_2(k)) = -1$. The dynamics of states $x_1(k)$ and $x_2(k)$ of the controlled discrete-time nonlinear dynamic system and corresponding controls $u_1(k)$ and $u_2(k)$ obtained by solving two difference equations indicated above are shown in Fig. 4. As one can see in this figure, the trajectories of state variables converge to values equal to zero during a few discrete-time steps that is almost by one order less than that obtained for the same data in [37]. Moreover, the method described above does not need using any approximation unlike the neural dynamic programming techniques that needs a NN approximation of actor and critic. In addition, the approach given in [37] requires solving HJB equation in real time and provides only nearly optimal control problem solution. The method presented above gives the possibility to solve optimal control problem, increase accuracy of the obtained solution, reduce its time complexity, and simplify software and hardware implementation of the controller. Note that despite that $g_1(x_1(k)) = 0$ and the first difference equation of (1) indicated above is the partial case of the simplified difference Eq. (2), the trajectories of state variables converge to the correct zero steady states.

Example 3 Let us apply the approach presented above to the optimal control of permanent-magnet motor of linear

Fig. 3 Dynamics of states $x_1(k), x_2(k), x_3(k), x_4(k)$ and $x_5(k)$ of the discrete-time controlled nonlinear dynamic systems described by the set of difference Eqs. (16) with noise, gradient trajectories $\Delta V/\Delta x_1, \Delta V/\Delta x_2, \Delta V/\Delta x_3, \Delta V/\Delta x_4$ and $\Delta V/\Delta x_5$ of value function (17), and dynamics of control functions $u_1(k), u_2(k), u_3(k), u_4(k)$ and $u_5(k)$ (18)

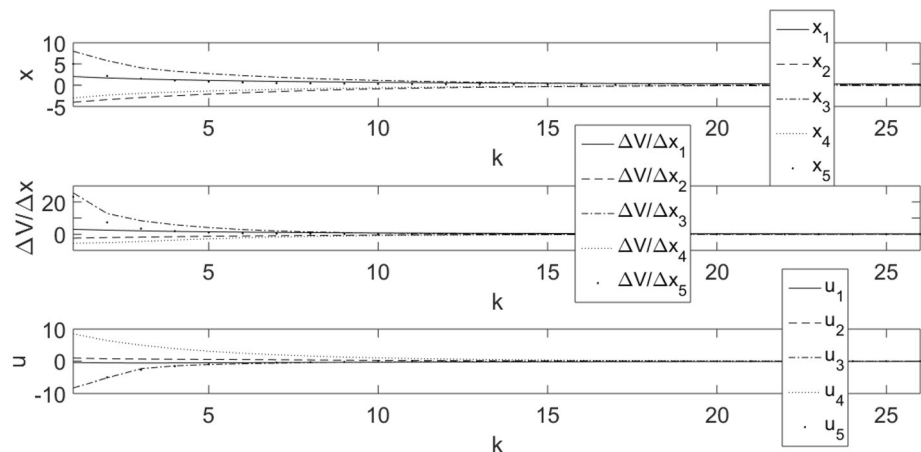
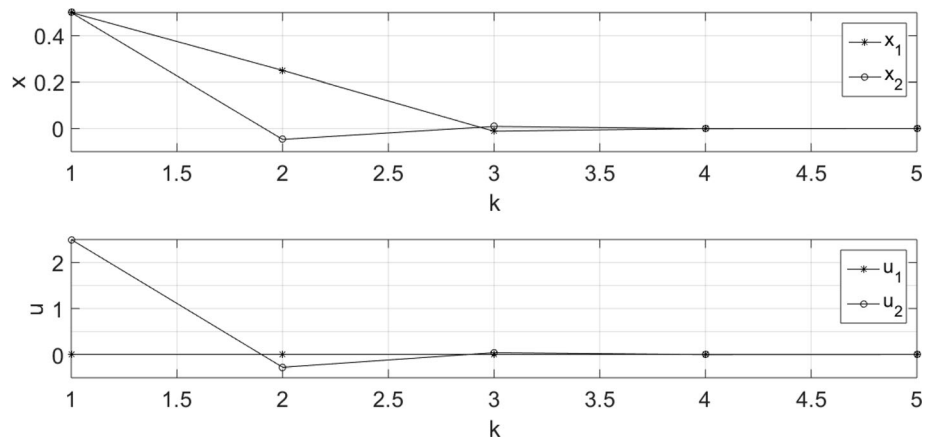


Fig. 4 The trajectories of system state variables $x_1(k)$ and $x_2(k)$, and corresponding controls $u_1(k)$ and $u_2(k)$



type used for accurate positioning adopted from [11]. The difference equation describing a positioning of the motor is given by

$$x_2(k+1) = x_2(k) + \tau(-6.49x_2(k) - 1.54u(k) + 0.19f_f + 0.19f_r),$$

where $x_2(k) = (x_1(k+1) - x_1(k))/\tau$, $x_1(k)$ is the k th discrete value of motor position, $u(k)$ is the k th discrete value of motor control, $f_f(k) = (f_c + f_v|x_2(k)|)\text{sgn}(x_2(k))$ is the k th discrete value of frictional force, $f_c = 3\text{N}$ is the Coulomb friction, $f_v = 10\text{N}$ is the viscous friction, $f_r(k) = A_r \sin(2\pi x_1(k)/T_r)$ is the k th discrete value of force ripple, $A_r = 3\text{N}$, $T_r = 7.12 \times 10^{-2}$ m. A k th discrete value of needed trajectory of the motor, in other words, trajectory of reference is taken as $r(k) = A \sin(2\pi f k \tau)$, where $A = 0.08$ m, $f = 0.25$ Hz.

In order to describe the motor position, we use the difference Eq. (1), where

$$f(x(k)) = \begin{bmatrix} x_2(k) \\ -6.5x_2(k) - (0.56 + 1.85|x_2(k)|)\text{sgn}(x_2(k)) \\ -0.56 \sin(28.09\pi x_1(k)) \end{bmatrix},$$

$$g(x(k)) = \begin{bmatrix} 0 \\ -1 \end{bmatrix}, \quad x(k) = (x_1(k), x_2(k))^T, \quad m = n = 2.$$

Specifically, the motor position $x_1(k)$ is described by the first difference equation $x_1(k+1) = x_1(k) + \tau x_2(k)$ that is a special case of (2) since $g(x_1(k)) = 0$. We consider such difference equation as one-dimensional with only one state variable $x_1(k)$ since values of $x_2(k)$ are determined from the second difference equation in each discrete-time point k . The second difference equation is also considered as one-dimensional with only one state variable $x_2(k)$ since values of $x_1(k)$ are determined from the first difference equation in each discrete-time point k . The second difference equation is presented as follows:

$$x_2(k+1) = x_2(k) - \tau \gamma_2 \sqrt{(f_2(x_2(k)))^2 + R^{-1}Q(x_2(k))(g_2(x_2(k)))^2 S(x_2(k))}, \quad \text{where}$$

$f_2(x_2(k)) = -6.5x_2(k) - (0.56 + 1.85|x_2(k)|)\text{sgn}(x_2(k)) - 0.56 \sin(28.09\pi x_1(k))$, $g_2(x_2(k)) = 2.5$, $\gamma_2 = 2.0$, $R = 1.5$, $Q(x_2(k)) = 0.5$. We add the normally distributed random noise with zero mean and standard deviation 1×10^{-5} to the right-hand side of the second difference equation to simulate the controller operation in the presence of disturbances of its nonlinearities. It is assumed that the condition $|r_2(k+1) - r_2(k)| < |y_2(k+1) - y_2(k)|$ is satisfied in this example. For this reason, to track the reference $r(k)$, we replace $x_1(k)$ with $e_1(k) = y_1(k) - r_1(k)$ in the first difference equation, where $y_1(k)$ is the first augmented discrete-time state variable, $e_1(k)$ is the first discrete-time tracking error. On the other hand, $x_2(k)$ is replaced with $e_2(k) = y_2(k) - r_2(k)$ in the second difference equation, where $y_2(k)$ is the second discrete-time state variable, $e_2(k)$ is the second discrete-time tracking error.

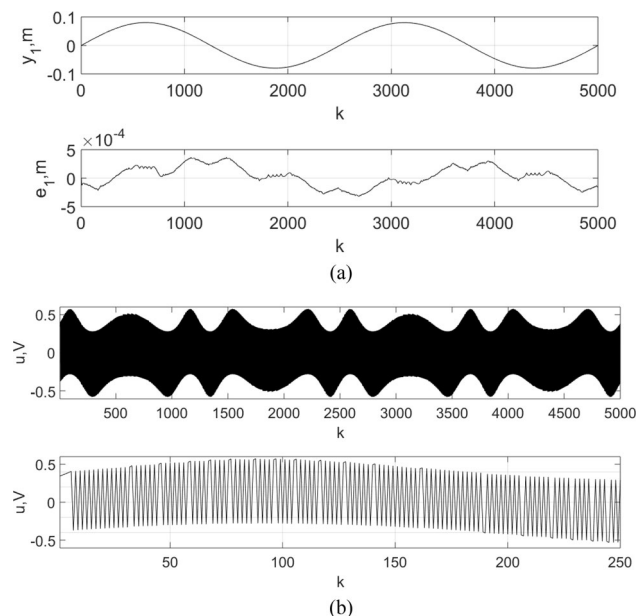
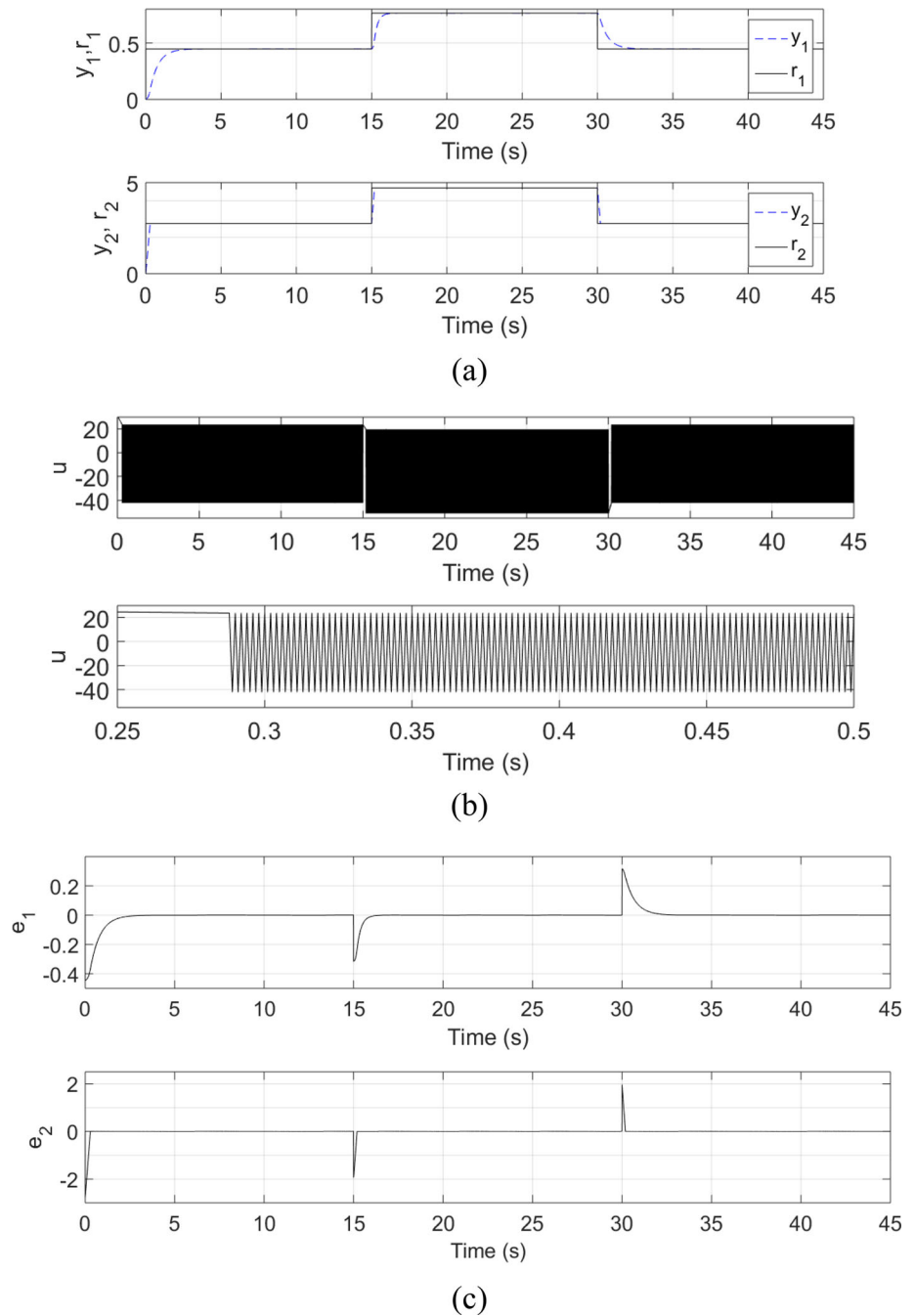


Fig. 5 **a** dynamics of the motor position $y_1(k)$ and tracking error $e(k)$, **b** dynamics of the control function $u(k)$

Fig. 6 **a** trajectories of augmented state variables $y_1(t), y_2(t)$ and trajectories of references $r_1(t), r_2(t)$ **b** trajectories of control $u(t)$ **c** trajectories of references $e_1(t)$ and $e_2(t)$



We set $\tau = 1.6 \times 10^{-3}$ and initial values of the state variables $y_1(1) = 0$ and $y_1(1) = 0.1$. The dynamics of $y_1(k)$, $e_1(k)$ and $u(k)$ are presented in Fig. 5. As one can see, $y_1(k)$ tracks $r_1(k)$ with small errors and $u(k)$ is changed within a bounded range. Note that in this example $23.244 \leq |y_2(k+1) - y_2(k)|/|r_2(k+1) - r_2(k)| \leq 2.2055^4$. Therefore, the condition $|r_2(k+1) - r_2(k)| < < |y_2(k+1) - y_2(k)|$ is satisfied in this example. In contrast to the PID controller described by integral-differential

equation in [11], the controller presented above is simpler since it is described by algebraic equation while demonstrating high accuracy of its operation. This allows to simplify a software and hardware implementation of the controller and increase the speed of its operation.

Example 4 We apply the method presented above to the optimal control of nonlinear cooling continuous stirred tank reactor adopted from [38]. The dynamics of the

reactor can be described in normalized form by the nonlinear differential Eq. (1), where

$$f = \begin{bmatrix} x_1(k) + \tau \left(-\alpha x_1(k) + D(1 - x_1(k))e^{\frac{x_2(k)}{1+x_2(k)}} \right) \\ x_2(k) + \tau \left(-\alpha x_2(k) + BD(1 - x_1(k))e^{\frac{x_2(k)}{1+x_2(k)}} - \beta x_2(k) \right) \end{bmatrix}, \quad g = \begin{bmatrix} 0 \\ \beta \end{bmatrix},$$

$\alpha = 1.0, \quad \beta = 0.3, \gamma = 20.0, \quad B = 1.0, \quad D = 0.072, \quad \tau = 0.001, x_1(1) = x_2(1) = 0.$ The objective is to track

the reference trajectories $r_1(t) =$

$$\begin{cases} 0.4472, & \text{if } 0 \leq t \leq 15; \\ 0.7646, & \text{if } 15 < t < 30; \\ 0.4472, & \text{if } 30 \leq t < 45 \end{cases} \quad \text{and} \quad r_2(t) = \begin{cases} 2.7520, & \text{if } 0 \leq t \leq 15; \\ 4.7052, & \text{if } 15 < t < 30; \\ 2.7520, & \text{if } 30 \leq t < 45. \end{cases}$$

We assume that the condition $|r_i(k+1) - r_i(k)| < |y_i(k+1) - y_i(k)|$, where $i = 1, 2$ is met in this example. Therefore, in order to track the reference $r_1(t)$, we replace $x_1(k)$ with $e_1(k) = y_1(k) - r_1(k)$ in the first difference equation, where $y_1(k)$ is the first discrete-time state variable, $e_1(k)$ is the first discrete-time tracking error. On the other hand, $x_2(k)$ is replaced with $e_2(k) = y_2(k) - r_2(k)$ in the second difference equation, where $y_2(k)$ is the second discrete-time state variable, $e_2(k)$ is the second discrete-time tracking error. The trajectories of augmented state variables $y_1(t)$ and $y_2(t)$, and references $r_1(t)$ and $r_2(t)$ are shown in Fig. 6a. Figure 6b presents the dynamics of control $u(t)$. Figure 6c demonstrates the trajectories of references $e_1(t)$ and $e_2(t)$.

As it can be seen in Fig. 6a–c, the augmented state variables $y_1(t)$ and $y_2(t)$ track the reference trajectories $r_1(t)$ and $r_2(t)$ with the accuracy close to that obtained in [38]. The control $u(t)$ and references $e_1(t)$ and $e_2(t)$ are changed in limited ranges. Note that in this example $1.8641 \cdot 10^{-9} \leq |y_1(k+1) - y_1(k)| \leq 0.9012 \cdot 10^{-4}$ and $0.0095 \leq |y_2(k+1) - y_2(k)| \leq 0.0106$ while $|r_1(k+1) - r_1(k)| = 0$ and $|r_2(k+1) - r_2(k)| = 0$ if $0 \leq t < 15$, $15 < t < 30$, and $30 < t < 45$. Therefore, the condition $|r_i(k+1) - r_i(k)| < |y_i(k+1) - y_i(k)|$, $i = 1, 2$ holds in this example. Unlike the approach described above, the method presented in [38] requires solving minimization problem at each iteration by recurrent general projection NN to obtain optimal control. Since our approach is related to obtaining optimal control

analytically, this reduces the solution computational complexity of the optimal tracking control problem.

6 Conclusions

Optimal control of NN type for the affine in the input discrete-time nonlinear dynamic systems with known internal dynamics has been presented. The control is described by variable structure algebraic equation derived analytically. A functional block diagram of the controlled nonlinear dynamic system is given. Implementation aspects of the controller in software and hardware are discussed. The existence and uniqueness of steady states of the controlled nonlinear dynamic system has been proven. Global asymptotic stability and convergence of state variable trajectories of the controlled systems to steady states has been proven. It is shown that the number of time steps needed for controlled system state variable trajectories to reach the neighborhood of steady states is finite. Analysis of sliding modes of the controller operation has been performed. The controller operation in the presence of disturbances of its nonlinearities has been analyzed. Application of the controller for a special case of optimal tracking control has been discussed. Computer simulation examples of the controller operation illustrating its high efficiency have been provided. Comparatively to other analog, the controller has lower computational complexity and does not need any training. Therefore, it can be concluded that the controller can be used for efficient optimal control of affine in the input, time-invariant discrete-time nonlinear dynamic systems with the known internal dynamics.

Data availability All data generated and analyzed during this study are included in this published article.

Declarations

Conflict of interest The author has no relevant financial or non-financial interests to disclose.

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